

Online / AI Recognition Prototype

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Overview & Purpose

This is an **online, AI-supported Wellbotics prototype** that explores how **digital systems can recognize, interpret, and respond to students' answers**—particularly in relation to **social–emotional learning (SEL)**. Unlike the Bee-Bot Garden Prototype, which centers on **embodied physical interaction**, this section investigates how **recognition, interpretation, and feedback** can occur within a **screen-based environment**.

This prototype does not aim to replace physical robotics experiences. Instead, it explores a **complementary modality**: how AI systems might support **reflection, assessment, and emotional awareness** when learner input is verbal, written, or selected rather than enacted through movement.

Core Question

The section is explicitly framed around a guiding inquiry: **“What can AI do?”**

From this question, the prototype explores how AI might:

- Recognize learner responses
- Interpret meaning (both cognitive and emotional)
- Support evaluation and feedback
- Scaffold SEL reflection

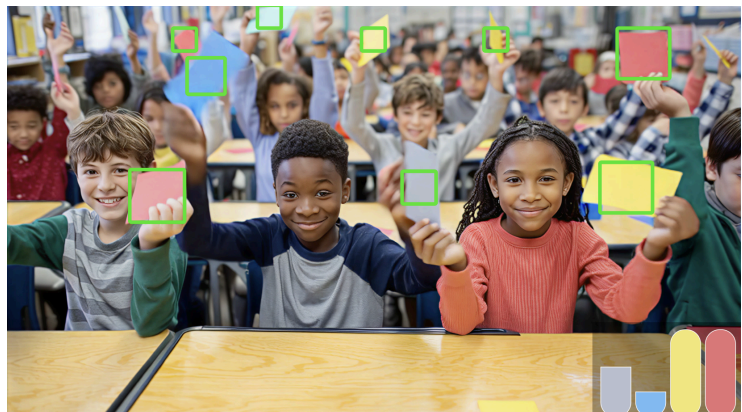
Rather than defining a full game system, this prototype focuses on **capability exploration**—similar in spirit to the Ozobot section, but oriented toward **interpretive intelligence** rather than robotic affordances.

Core Functional Focus: Recognition

The defining feature of this prototype is **recognition**.

The system is designed to:

- Identify what a student is responding to
- Recognize patterns in student answers



- Distinguish between different emotional or conceptual states reflected in responses
- Support facilitators and learners by making responses legible and actionable

Importantly, the recognition focus extends beyond correctness. The prototype explicitly considers **emotional and reflective dimensions**, positioning AI as a tool for understanding how students are engaging, not just what they answer.

Learning Domains Addressed

This prototype aligns the online prototype with the same core learning goals as the physical systems, but through a different interaction mode.

1. Social–Emotional Learning (SEL)

The prototype emphasizes AI's potential to:

- **Recognize emotion** in student responses
- Support **emotion management** through feedback
- Enable reflective learning moments

By interpreting student answers, the system can surface:

- Emotional states (e.g., confusion, confidence, frustration)
- Behavioral patterns
- Opportunities for reflection or guided discussion

This aligns with the Wellbotics emphasis on SEL as an **ongoing process**, not a one-time assessment.

2. Reflection & Interpretation

Unlike the Bee-Bot prototype—where reflection is triggered by physical game states—this system uses **AI-mediated interpretation** to prompt reflection.

Examples of reflective support include:

- Helping students notice patterns in their own responses
- Encouraging them to articulate reasoning or feelings
- Providing structured feedback that links choices to outcomes

The system positions AI not as an authority, but as a **mirror and guide**.

Relationship to Programming and Systems Thinking

While this prototype does not emphasize direct robot programming, it still supports **systems thinking** by:

- Making cause-and-effect relationships visible
- Helping learners connect inputs (answers, choices) to outputs (feedback, interpretation)

- Encouraging metacognition (“Why did I respond this way?”)

In this sense, the online prototype abstracts the same principles found in the physical game:

- Actions → system response → reflection → adjustment

Role of the Facilitator

Although the system is digital, the facilitator remains important.

The AI system is designed to:

- **Support**, not replace, facilitators
- Surface insights that facilitators can respond to
- Provide a shared reference point for discussion

This preserves Wellbotics’ core philosophy: learning is **relational and guided**, even when mediated by technology.

Status

This prototype is clearly **conceptual and exploratory**, rather than fully specified.

What is defined:

- Purpose (recognition, interpretation, feedback)
- Learning focus (SEL, reflection)
- Role of AI (supportive, interpretive)

What is intentionally left open:

- Specific interaction mechanics
- Interface details
- Assessment logic
- Degree of automation vs facilitator control

This openness reflects the prototype’s role as a **design probe**, not a finalized product.

Conclusion

The Online / AI Recognition Prototype extends Wellbotics’ core philosophy into the digital domain by investigating how **AI can help make student thinking and feeling visible**. Rather than focusing on automation or efficiency, the system emphasizes **interpretation, reflection, and emotional awareness**, aligning closely with SEL goals.

By treating AI as a **reflective partner** rather than a judge, this prototype reinforces Wellbotics’ central insight: learning—whether physical or digital—is most powerful when systems help learners understand **the relationship between their actions, emotions, and outcomes**.

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